Appendix B
AI/ML Pipeline

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This appendix presents the complete AI/ML pipeline specification for the HELIOS autonomous solar-storm warning system. Section B.1 defines the neural network architecture and multi-task learning formulation. Section B.2 documents the training methodology and synthetic dataset generation. Section B.3 derives the dose-based severity classification framework aligned with NASA-STD-3001. Section B.4 specifies the onboard ensemble consensus mechanism and Triple Modular Redundancy (TMR) decision logic. Section B.5 reports MVP validation results against historical CME events.

B.1 Neural Network Architecture

B.1.1 Dual-Head Model Design

The HELIOS magnetic structure inference model employs dual-head neural network architecture for simultaneous Bz regression and severity classification. A shared encoder extracts latent representations from a 16-dimensional input feature vector (Table B.1), which are then processed by two task-specific output heads. Multi-task learning enables joint optimization: the shared encoder captures CME physical structure common to both tasks, while dedicated heads specialize in their respective objectives.

The shared encoder consists of four fully connected layers with dimensions [16 → 128 → 256 → 128 → 64], each followed by Layer Normalization, GELU activation, and Dropout (p = 0.2). The Bz regression head maps the 64-dimensional encoded representation through a hidden layer (64 → 32) to a two-element output: predicted mean μ and log-variance $log(\sigma^2)$. This heteroscedastic formulation enables input-dependent uncertainty estimation — the model learns to express higher uncertainty for ambiguous CME configurations. The severity classification head applies an identical hidden layer (64 → 32) followed by a 4-class output layer with softmax activation, producing probability distributions over Low, Moderate, High, and Extreme severity classes. The complete model contains approximately 82,000 trainable parameters.

B.1.2 Loss Function

Training minimizes a weighted combination of two losses. The Bz regression head uses a heteroscedastic loss function that penalizes both prediction error and overconfident uncertainty estimates:

$$\mathcal{L}\_Bz = (1/N)\,\Sigma\,[\,½ \cdot exp(-\hat{\sigma}^2) \cdot (\hat{\mu} - y)^2 + ½ \cdot \hat{\sigma}^2\,]$$

where $\hat{\mu}$ is predicted Bz mean, $\hat{\sigma}^2$ is the predicted log-variance, and y is the ground-truth Bz value. The first term penalizes prediction error inversely weighted by learned uncertainty; the second regularization term prevents the trivial solution of infinite uncertainty. The severity head uses weighted cross-entropy loss with class weights [1.0, 1.5, 2.5, 5.0] to address class imbalance and emphasize correct classification of high-severity events. The total loss combines both objectives:

$$\mathcal{L}\_total = \alpha \cdot \mathcal{L}\_Bz + \beta \cdot \mathcal{L}\_severity$$

The loss weighting (α = 0.5, β = 0.5) balances Bz regression accuracy with severity classification performance during training. At inference time, severity is independently derived from the predicted Bz value through the deterministic dosimetry framework (Section B.3).

#	Feature	Units	Bounds	Source
1	cme_speed	km/s	[300, 3500]	Coronagraph kinematics
2	angular_width	deg	[15, 360]	Coronagraph morphology
3	source_latitude	deg	[−90, 90]	Flare location
4	source_longitude	deg	[−180, 180]	Flare location
5	expansion_rate	R⊙/h	[0.1, 5.0]	Coronagraph evolution
6	acceleration	m/s^2	[−500, 500]	Coronagraph kinematics
7	L1_viewing_angle	deg	[0, 180]	Constellation geometry
8	L4_viewing_angle	deg	[0, 180]	Constellation geometry
9	L5_viewing_angle	deg	[0, 180]	Constellation geometry
10	brightness_asymmetry	ratio	[0.1, 10.0]	Coronagraph photometry
11	parallax_L1L4	R⊙	[0, 50]	Stereoscopic triangulation
12	parallax_L1L5	R⊙	[0, 50]	Stereoscopic triangulation
13	parallax_L4L5	R⊙	[0, 50]	Stereoscopic triangulation
14	detection_time	hours	[0.1, 24]	Post-eruption timing
15	triangulation_quality	score	[0, 1]	Geometric quality metric
16	observation_completeness	score	[0, 1]	Data availability metric

Table B.1 — 16-dimensional input feature vector. Features 1–6 encode intrinsic CME properties derived from coronagraph observations; features 7–9 capture HELIOS constellation geometry; features 10–13 encode stereoscopic triangulation measurements; features 14–16 characterize observation quality. All features are normalized to [0, 1] via predefined bounds for model input.

B.2 Training Methodology

B.2.1 Dataset Composition

The training dataset combines synthetic and historical CME events to achieve both volume and physical fidelity. A total of 10,000 synthetic events were generated using the flux rope physics model described below, with stratified sampling to ensure representation across all severity classes (35% Low, 30% Moderate, 20% High, 15% Extreme). Nineteen well-characterized historical CME events (excluding Bastille Day 2000, reserved as unseen showcase)split into 12 training and 6 test events. Training events are oversampled 50x, yielding 600 historical training samples. Combined with 10,000 synthetic events: total = 10,600 training samples.

Six historical events were completely excluded from training and reserved as an independent test set: the Halloween Storm 2 (October 29, 2003), September 2017, January 2005, November 2001, May 2024, and October 2024 events. The Bastille Day 2000 event was separately excluded to serve as the primary validation showcase (Section B.5.2).

B.2.2 Bz Physics Model

Synthetic Bz ground-truth values are generated from a simplified flux rope model calibrated against historical observations:

$$Bz = -k \cdot \sqrt{}(v/1000) \cdot (w/60)^{3r10} \cdot |sin(\theta\_tilt)| + \varepsilon$$

where v is CME speed (km/s), w is angular width (degrees), θ_tilt is the flux rope tilt angle drawn from a uniform distribution U[−90°, 90°], and ε represents observational noise. The calibration constant k = 35.0 was tuned to reproduce the Bastille Day 2000 observation: v = 1674 km/s, w = 360° yields Bz ≈ −60 nT, consistent with the ACE L1 measurement. The exponent 0.3 on angular width captures the empirical relationship between CME lateral extent and geo-effectiveness.

B.2.3 Training Configuration

The model was trained with the Adam optimizer (learning rate 5×10^{-4}, weight decay 10^{-5}) using a batch size of 64 over a maximum of 100 epochs. Early stopping with patience of 20 epochs prevented overfitting, monitored on the validation loss. A ReduceLROnPlateau scheduler (factor 0.5, patience 5) adapted the learning rate during training. Gradient clipping (max norm 1.0) ensured training stability. Input features were normalized to [0, 1] via min-max scaling using predefined FEATURE_BOUNDS (Table B.1); Bz targets were linearly scaled to [0, 1] within the range [−80, 0] nT.

B.3 Severity Classification Framework

B.3.1 Dosimetry Model

Severity classification is performed through a deterministic dosimetry framework applied to the neural network's Bz prediction — it is not a neural network output. Classification operates on predicted radiation dose, not on Bz ranges directly, because the Bz component serves as an empirical proxy for CME magnetic energy content rather than a direct radiation driver. In deep space, without magnetospheric shielding, astronauts receive the full solar energetic particle flux. The dose estimation formula is:

$$D\,(mSv) = K \times |Bz|^{\alpha} \times \sqrt{v_{CME}} \times t_{exposure}$$

where D is the estimated dose in millisieverts (mSv), Bz is the predicted southward interplanetary magnetic field component (nT), v is CME arrival speed (km/s), and t is exposure duration (hours). The coefficient K = 0.0132 represents the median calibration across eight historical solar particle event reconstructions [1, 2, 3]: August 1972, Bastille Day 2000, Halloween Storms 2003, January 2005, March 1989, September 2017, May 2024, and the Carrington 1859 proxy. The Bz–energy scaling exponent α = 1.3 follows empirical relationships established in the geomagnetic storm literature [4]. Exposure duration t = 10 hours represents a pessimistic worst-case EVA scenario.

B.3.2 Severity Thresholds

Class	Dose Range (mSv)	NASA SPE Limit	Crew Response	Bz 800 km/s
Low (0)	10-50	4-20%	Enhanced monitoring	-7 to -5 nT
Moderate (1)	50-100	20-40%	Shelter advisory	-13 to -7 nT
High (2)	100-250	40-100%	Mandatory shelter	-25 to -13 nT
Extreme (3)	>250	>100%	EVA abort; emergency shielding	< -25 nT

Table B.2 — Dose-based severity classification. Bz values are illustrative at a reference speed of 800 km/s; operational classification uses dose thresholds exclusively. The Extreme threshold (250 mSv) is aligned with NASA-STD-3001 Vol. 1, Rev. C requirement [V1 4031] [5] for single solar particle event effective dose limits.

B.3.3 NASA-STD-3001 Compliance

The severity framework is designed around the dose limits specified in NASA-STD-3001 Volume 1, Revision C (2023) [5]. The SPE effective dose limit of 250 mSv per event (requirement [V1 4031]) defines the boundary between High and Extreme severity classes. The career effective dose limit of 600 mSv (requirement [V1 4030]) and the ALARA (As Low As Reasonably Achievable) principle (requirement [V1 4029]) are reflected in the graduated response protocol, which initiates protective actions well below the regulatory limit.

B.4 Consensus and Decision Architecture

The HELIOS warning pipeline separates probabilistic AI inference from deterministic safety decisions through two distinct processing layers. The onboard ensemble consensus (Section B.4.1) performs epistemic consistency evaluation across three independent ML predictions from three onboard neural network models, producing a single best-estimate Bz value with calibrated confidence through tiered agreement analysis. The TMR decision layer (Section B.4.2) then processes this consensus output through redundant deterministic logic blocks with majority voting to generate fault-tolerant crew warnings.

B.4.1 Onboard Ensemble Consensus

Each constellation node (L1, L4, L5) operates three independent neural network models that process locally acquired coronagraph imagery and in-situ magnetic field, plasma, and energetic particle measurements. Each model independently constructs the 16-dimensional feature vector (Table B.1) from the satellite's observational data, processes it through its own trained weights, and produces a separate Bz prediction with heteroscedastic uncertainty and severity class probabilities. The three models are architecturally identical but trained with different random initializations, ensuring epistemic diversity in their predictions. The onboard consensus module receives the three independent predictions from these co-located models and performs epistemic consistency evaluation entirely within the satellite's processing unit — no inter-satellite transmission of ML predictions is required for the consensus step. Inter-satellite optical links (8.3-minute one-way light time between L1 and L4/L5) serve the separate function of exchanging raw observational data and finalized warning products, not intermediate ML outputs.

Epistemic consistency evaluation follows a tiered agreement protocol. If all three predictions yield the same severity class (3/3 exact match), the system achieves FULL_FUSION status with the arithmetic mean of all Bz predictions as the consensus value and proceeds directly to the TMR decision layer. If two of three predictions agree (2/3 agreement), the system enters EXTENDED_EVENT_ANALYSIS — a procedural evaluation rather than a simple majority vote. If fewer than two predictions agree, the system enters ABORT status immediately, using the median Bz as a robust fallback, logging all diagnostic data, and triggering a downlink for ground review.

Extended Event Analysis procedure. When two of three onboard AI models produce consistent predictions (2/3 agreement on severity class), the Extended Event Analysis procedure is initiated rather than accepting the agreeing pair at face value. This distinction is critical: two models may agree due to shared sensitivity to particular input features or correlated training-data biases, and a simple majority vote would not detect such systematic errors. The procedure consists of two steps:

(1) Subset fusion: The agreeing pair's Bz predictions and severity probabilities are fused into a single estimate using weighted averaging, where the weights are inversely proportional to each prediction's reported uncertainty ($\hat{\sigma}^2$ from the heteroscedastic head). The dissenting model's prediction is retained but excluded from the fused estimate. (2) Physical plausibility check: The fused estimate is validated against physics-based constraints derived from the CME's observed properties — specifically, the empirical Bz–speed correlation from the dosimetry model (Equation

B.4), the expected Bz range given the CME's angular width and source location, and consistency between the triangulated 3D trajectory and the predicted magnetic structure orientation.

If the plausibility check passes, the system confirms the 2/3 result and proceeds as PARTIAL_FUSED: the subset-fused Bz estimate is forwarded to the TMR decision layer with a reduced confidence flag and a log entry documenting the dissenting prediction for post-event analysis. If the plausibility check fails — indicating that the agreeing pair's prediction is physically inconsistent despite their nominal agreement — the system downgrades the consensus to 1/3 equivalent and enters the ABORT sequence: a conservative warning is issued based on the median Bz, all raw predictions are downlinked, and the system triggers a ground review before the next inference cycle. Before reaching TMR, both FULL_FUSION and PARTIAL_FUSED outputs are augmented with consensus data from other constellation nodes via real-time inter-satellite exchange, where available. Table B.3 summarizes all consensus evaluation outcomes.

Vote	Severity Agreement	Procedure	Status	Action
3/3	Exact match	Full fusion (mean of 3)	FULL_FUSION	Proceed to TMR decision layer
2/3	Two agree	Extended Event Analysis: subset fusion + plausibility check	EXTENDED_EVENT_ANALYSIS	If pass → PARTIAL_FUSED (reduced confidence); if fail → downgrade to ABORT
<2/3	Divergent	No fusion possible	ABORT	Median Bz fallback; downlink + ground review

Table B.3 — Ensemble consensus evaluation outcomes. Bz tolerance: ±10 nT; severity tolerance: ±1 class. Extended Event Analysis applies subset fusion and physics-based validation before accepting 2/3 agreement.

MVP simulation. The MVP employs a three-seed ensemble (seeds 42, 123, 456) of independently trained models to approximate the epistemic diversity expected from three separately initialized onboard models. Each ensemble member uses identical architecture and training data but different random initialization, producing genuinely independent learned representations. For severity classification, the MVP uses simplified Bz-threshold boundaries (Extreme: Bz ≤ −30 nT; High: −30 < Bz ≤ −20 nT; Moderate: −20 < Bz ≤ −10 nT; Low: Bz > −10 nT) to maintain consistency between training labels and evaluation. The operational system will apply the full dose-based classification (Section B.3.2). At inference, severity probabilities are derived by integrating the Gaussian predictive distribution N(μ_Bz, σ^2) over each threshold interval, providing principled confidence estimates. The MVP implements the consensus agreement tiers (3/3, 2/3, <2/3) and status classification, but simplifies the Extended Event Analysis procedure: subset fusion uses unweighted averaging of the agreeing pair. The full physics-based plausibility validation is deferred to the operational multi-model deployment.

B.4.2 TMR Decision Layer

The decision layer implements Triple Modular Redundancy (TMR) for fault-tolerant warning issuance. Three identical processing blocks receive the same consensus input — the fused Bz estimate, CME velocity, and geometric trajectory from the ensemble layer — and execute a deterministic threshold evaluation in parallel:

(1) Calculate radiation dose via Equation B.4; (2) Map dose to severity class via Table B.2 thresholds; (3) Generate a boolean GO/NO-GO output.

A majority voter compares the three boolean outputs — this is the only layer in the HELIOS architecture that employs true majority voting, as it operates on identical deterministic computations from identical inputs. With 3/3 agreement, the warning is issued at nominal confidence. With 2/3 agreement (indicating a probable single-module hardware fault), the warning is issued with the uncertainty flagged and a ground review triggered. With fewer than 2/3 in agreement, the warning is suppressed and mandatory manual intervention is initiated. TMR protects against hardware-level anomalies (single-event upsets, transient faults, radiation-induced bit flips) rather than algorithm errors — under nominal hardware conditions, all three blocks produce identical outputs from identical inputs. This is standard practice for safety-critical aerospace systems (Space Shuttle GPC, ISS C&DH, F-35 flight control).

The architectural separation between probabilistic AI inference (ensemble consensus) and deterministic decision logic (TMR voting) is deliberate. AI produces the best available state estimate; the safety decision is made by well-understood, formally verifiable threshold logic. This design ensures that neural network uncertainty never directly controls crew safety actions.

B.5 MVP Validation Results

B.5.1 Test Set Performance

Model performance was evaluated on six historical CME events completely excluded from training. Across the test set, the pipeline achieved a mean absolute error (MAE) of 6.3 ± 5.6 nT for Bz prediction, representing a 36% improvement over a geometric baseline estimate of 9.9 nT. Severity classification accuracy was 83.3% (5 of 6 events correct), with 100% adjacent-class accuracy — the single misclassification (January 2005: High predicted as Extreme) represents a conservative failure mode that overestimates severity, maintaining crew safety margins. Table B.4 presents the complete test set results.

Event	Date	True Bz (nT)	Pred Bz (nT)	Error (nT)	True Sev	Pred Sev	Correct
Halloween Storm 2	2003	−49.0	−55.7	6.7	Extreme	Extreme	✓
September 2017	2017	−32.0	−31.5	0.5	Extreme	Extreme	✓
January 2005	2005	−25.0	−43.9	18.9	High	Extreme	✗
November 2001	2001	−30.0	−35.2	5.2	Extreme	Extreme	✓
May 2024	2024	−50.0	−55.5	5.5	Extreme	Extreme	✓
October 2024	2024	−38.0	−35.5	2.5	Extreme	Extreme	✓
Aggregate (N = 6)				**6.3 ± 5.6**		**83.3%**	**5/6**

Table B.4 — Independent test set validation on six unseen historical CME events. All events were completely excluded from training. January 2005 misclassification (High → Extreme) represents a conservative error that overestimates severity. Aggregate MAE = 6.3 ± 5.6 nT; improvement over geometric baseline: 36%.

B.5.2 Bastille Day 2000 End-to-End Validation

The Bastille Day 2000 event (X5.7 flare, 1674 km/s full-halo CME, source region N22W07) was used as the primary end-to-end validation showcase. This event was completely excluded from the training set. Ground truth Bz = −60.0 nT was measured by the ACE spacecraft at L1. Three independently trained ensemble members (seeds 42, 123, 456) produced predictions of −55.2, −56.0, and −53.3 nT for the three simulated onboard models respectively. The ensemble consensus yielded Bz = −54.9 ± 1.1 nT (inter-model spread: 2.7 nT) with 3/3 FULL_FUSION status and a prediction error of 5.2 nT (8.6% relative), meeting the operational target of < 10 nT.

All three inference passes classified the event as Extreme with confidence ranging from 90.7% to 92.8% (severity class 3, unanimous). The physical model cross-validation applied the consensus Bz to Equation B.4: $D = 0.0132 \times |54.9|\hat{}1.3 \times \sqrt{1674} \times 10 = 985\, mSv$, corresponding to 394% of the NASA-STD-3001 250 mSv SPE limit. The physical severity classification (Extreme) is consistent with the ML prediction, validating the end-to-end pipeline from neural network inference through dosimetry to crew warning generation. Table B.5 summarizes the validation results.

Component	Value	Status
Ensemble Predictions (Model 1, 2, 3)	−55.2, −56.0, −53.3 nT	Inter-model spread: 2.7 nT
Consensus Bz	−54.9 ± 1.1 nT	3/3 FULL_FUSION
Mean Absolute Error	5.2 nT (8.6%)	< 10 nT target ✓
Severity Classification	Extreme (class 3)	Confidence: 91–93%
Physical Model Dose	985 mSv	394% of NASA SPE limit
ML-Physics Consistency	Extreme = Extreme	PASS ✓
Alert Level	RED — Critical	24.8 h to impact

Table B.5 — End-to-end MVP validation on the Bastille Day 2000 extreme CME event. Ground truth from ACE spacecraft (Bz = −60.0 nT). Predicted dose (985 mSv) exceeds the NASA-STD-3001 250 mSv SPE limit by a factor of 4, correctly triggering Extreme classification with a RED crew alert and seven recommended protective actions.

References

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